Performance Evaluation of Different Fuel Injection Struts for Scramjet Combustor

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Abstract. Strut-based fuel injection method substantially affects the combustion efficiency of the scramjet engine. The present research evaluates the combustion characteristics of different strut geometries by calculating the combustion efficiency (η_c) and total pressure loss (η_{ptl}) for improved performance of hydrogen fueled scramjet engine. Computational simulations are carried out by using the shear stress transport (SST) *k-ω* turbulence model. Firstly, the mesh independence study is conducted to choose the finest grid with results independent of the grid resolution. The methodology is then validated against the literature for the traditional DLR combustor geometry with single wedge strut, considering both reacting and non-reacting scenarios. Qualitative validation analysis is done by comparing the simulated density contours which show the shock-I and shock-II structures consistent with the experimental results. Static pressure variations along the top and bottom walls, as well as the centerline, align with experimental and computational results, validating the present study quantitatively. A thorough comparison between the performance of single wedge, lobed, and diamond strut are reported for a slightly modified geometry of traditional DLR combustor. It is noticed that the wedge and lobed strut designs create strong recirculation zones, yielding the highest η_c (69% for wedge, 67% for lobed strut), while the diamond strut, with no recirculation zone, shows the lowest η_c. The total pressure loss is highest for the lobed followed by wedge and diamond. This research provides a roadmap for researchers and industries exploring optimal fuel strut geometry for scramjet combustors.

Keywords: Hydrogen fuel, CFD, numerical simulation, ANSYS Fluent, multi-strut fuel injection method.

1 Introduction

In recent decades, numerous studies have been carried out, and significant efforts have been dedicated to enhancing the combustion efficiency of a scramjet combustor [1-7]. As compared to turbojet engines, scramjet engines can attain speeds ranging from Mach 4 to Mach 16. Although rocket engines are used for this flight regime, a scramjet provides a higher specific impulse and eliminates the need for an onboard oxidizer [8-10]. Inside the scramjet combustor, the freestream air propagates at supersonic speeds, causing the fuel to remain in the combustor for only 0.1

milliseconds. Therefore, actions need to be taken to slow down the supersonic flow within the combustor for high residence time. Several investigations have been carried out to increase the residence time through the recirculation zone created by the flame stabilization devices [11,12], so that the flame stays within the recirculation zone while being a source of ignition throughout the combustor length. Many researchers are increasingly focused on investigating scramjet combustion engines due to their numerous applications and advantages in aerospace and related fields, such as military missiles [10,13].

Inspired by the DLR scramjet model, Kummitha *et al.* [14] investigated various innovative and new strut designs for improved combustion characteristics of scramjet combustor. The study reported that the compressed air entering the combustion chamber is at a higher pressure compared to the basic DLR scramjet model, reducing ignition delay. Additionally, both the rocket and double arrow strut models increase boundary layer separation and vortex formation, thereby improving the mixing of fuel and air, ultimately leading to enhanced combustion efficiency. Kireeti *et al.* [15] solved Reynolds Averaged Navier-Stokes (RANS) equations using the Shear Stress Transport (SST) k-ω turbulence model to study the impact of flow physics and operating parameters such as injection pressure at 2, 4, 6, and 8 bars on a modified scramjet combustor. Authors validated the numerical methodology by analyzing the essential parameters such as wall pressures, shock wave patterns, and the static temperature of the combustor. It was observed that the shock wave moves upstream due to the pressure exerted by the main flow and with increasing H_2 jet pressure to 4 bar, a normal shock wave forms ahead of the strut leading to higher total pressure loss. The study reported that lower hydrogen jet pressures are more favorable for supersonic flows in the considered geometry. Additionally, as the injection pressure increases, the combustion zone expands over a larger area of the combustor, leading to a decrease in the combustion efficiency. Choubey *et al.* [16] recently carried out a study to examine the effect of two-strut fuel injection on the mixing efficiency and overall scramjet combustion performance, focusing on the flow field, flame lift-off characteristics, and combustion stabilization within the two-strut combustor. The simulation results indicated a more efficient distribution of combustion downstream leading to a wider range of temperature fluctuations for the two-strut configuration as compared to the single-strut configuration. Many studies have reported different strut designs and geometries to reduce the residence time and thus improve the mixing characteristics and combustion performance of a scramjet engine. However, a detailed and organized comparison between different strut designs, highlighting their specific advantages and applications has never been carried out before. Therefore, the present study aims to investigate several strut designs based on their mixing and combustion efficiency and total pressure loss.

The layout of the paper is as follows, in section 2, the adopted numerical methodology describing the governing equations and turbulence model employed to carry out the present research work is discussed. This is followed by sub-section 2.2 and 2.3 explaining the numerical schemes and boundary conditions. The mesh independent study is also presented in this section. Section 3 provides the qualitative and quantitative validation of the adopted numerical methodology and discusses the performance comparison of different strut configurations for non-reacting and reacting cases. Section 4 of the paper concludes with a summary and conclusions.

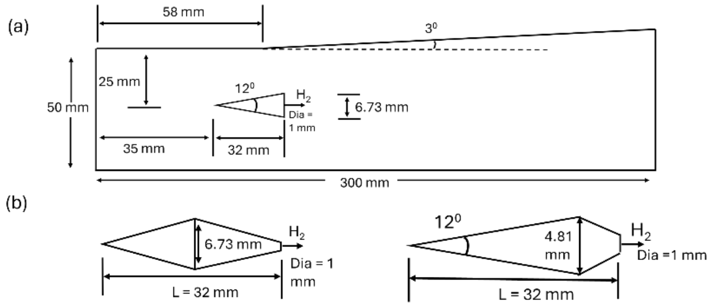

Fig. 1 The diagram illustrates (a) the single wedge strut-based DLR scramjet engine model, and (b) the lobed strut (left) and diamond strut (right).

2 Computational Domain

The schematic diagram of the traditional DLR (German Aerospace Centre) scramjet combustor with a single wedge strut positioned at x = 65 mm and equidistant from the combustor walls is shown in Figure 1a (all the dimensions are in mm). This single strut configuration is used for validation of the adopted numerical methodology. The strut injector has a length (L) of 32 mm, with a base height of 6.73 mm and the constant wedge angle (θ) is 12^0. Hydrogen is injected parallel into the airflow at Mach 1 via 1mm diameter (dia) hole, however air enters the combustor at $M = 2$. To examine the impact of different strut configurations on the mixing and combustion efficiency of the scramjet engine, three different types of struts; wedge, lobed and diamond, are considered. The schematic diagrams of lobed (left), diamond (right) struts are illustrated in Figure 1b. The length ($L = 32\ mm$) of the strut, wedge angle ($\theta = 12^0$) and diameter of hydrogen injection ($dia = 1\ mm$) are kept the same for all the three configurations of the struts. Furthermore, in the present analysis, a slightly modified scramjet combustor model is used with the length of the scramjet as 1000 mm and a divergence angle of 3^0 on both the top and bottom walls of the scramjet. The scramjet's height at the combustor entrance is 80 mm. Two-dimensional simulations of the DLR and modified scramjet combustor are conducted as the computational domain is axisymmetric.

2.1 Mesh Generation

An unstructured quadrilateral grid is employed for all the simulated cases using ANSYS Fluent 2024 R1. For better visualization, a coarse grid of 41000 points is shown in Figure 2. The simulated mesh for single wedge-strut based scramjet combustor comprises of 178472 grid elements and is finer near the strut and combustor walls to capture the sharp velocity gradients near the walls and near the centerline region to accurately resolve the turbulent mixing in the wake flow.

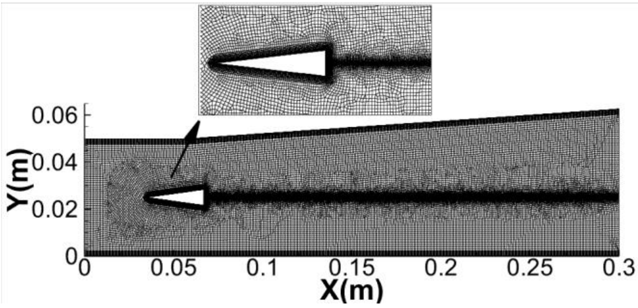

Fig. 2 Unstructured quadrilateral coarse mesh of single wedge strut-based scramjet engine.

2.2 Governing Equations

The two-dimensional Reynolds-Averaged Navier-Stokes (RANS) equations were solved in conjunction with species transport and the Shear Stress Transport (SST) k-ω turbulence model using the commercial CFD software Ansys Fluent 2024 R1. The governing equations are

Mass Conservation:

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x_i}(\rho u_i) = 0 \tag{1}$$

where ρ, u_i are density and *x*-component of velocity, respectively. Momentum conservation:

$$\frac{\partial}{\partial t}(\rho u_i) + \frac{\partial}{\partial x_j}(\rho u_i u_j) = -\frac{\partial P}{\partial x_i} + \frac{\partial \tau_{ij}}{\partial x_j} \tag{2}$$

where u_j is the *y*-component of velocity and P is the pressure term and τ_{ij} being the shear stress components. Energy Conservation:

$$\frac{\partial}{\partial t}(\rho E) + \frac{\partial}{\partial x_j}(\rho E u_j) = -\frac{\partial P u_j}{\partial x_i} + \frac{\partial(\tau_{ij} u_i)}{\partial x_i} + \frac{\partial}{\partial x_j}\left(k_{th}\frac{\partial T}{\partial x_j}\right) \tag{3}$$

Here, E is the total Internal Energy, and the last term constitutes the thermal diffusion effects. Since the fluid mixture is a thermally perfect gas, equation of state is used to solve the above equations. The ideal gas equation, $p = \rho RT$. The modeling of fuel-air mixing, and the movement of chemical species is described by the species transport or species conservation equation given by,

$$\frac{\partial(\rho Y_i)}{\partial t} + \frac{\partial}{\partial xj}(\rho Y_i u_j) = \frac{\partial}{\partial xj}\left(\rho D_{ij}\frac{\partial Y_i}{\partial x_j}\right) + R_i + S_i \tag{4}$$

Here, Y_i represents the mass fraction of species *i*, D_{ij} denotes the molecular diffusion term, R_i is the net rate of production of species *i* due to chemical reactions, and S_i is the chemical source term. For combustion modeling, a single-step reaction finite rate/eddy dissipation model is employed to capture the flow physics within the combustor. Hydrogen water reaction mechanism used is: $2H_2 + O_2 = 2H_2O$

2.3 Initial and Boundary Conditions

Here, air enters the scramjet combustor at supersonic Mach 2 with static pressure of 1 atm and fuel enters the combustor at sonic speed. All the variables are defined at the inflow using Dirichlet boundary conditions as tabulated in Table 1. On the injector and combustor walls, adiabatic, no-slip boundary conditions are defined, and at the outlet, Neumann boundary conditions are implemented. All the simulated cases use the incoming airflow conditions as the initial setup, assuming a steady-state flow.

Table 1. Initial Conditions

Parameter	Air	Hydrogen
Inlet Mach number	2.0	1.0
Inlet Velocity (U) (m/s)	1083.17	1200
Static Temperature (T) (K)	730	250
Static Pressure (P) (Pa)	1×10^5	1×10^5
Density (ρ) (kg/m^3)	1.002	0.097
Y_{O_2}	0.232	0
Y_{H_2O}	0.032	0
Y_{H_2}	0	1

2.4 Grid Convergence Study

A mesh independence study is conducted to select the finest grid, where further refinement does not change the computed result. Figure 3(a) shows that the static pressure profile along the top wall of the engine for three different grids with elements; 76381 (course), 92278 (medium) and 178472 (fine) mesh along with a zoomed-in view near the divergence angle (x = 58mm) is illustrated in the in-set. The variation of maximum value of static pressure on the lower wall with respect to different number of grid elements is shown in Fig. 3b. It is noticed that static pressure becomes constant for all the cases with grid elements more than 178472. Hence, a grid size of 178472 is used for subsequent analysis.

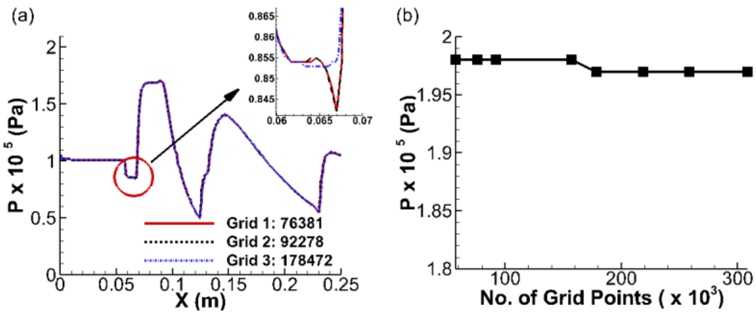

Fig. 3 (a) Variation of static pressure along the upper wall with zoomed-in view at x=58 mm (b) Maximum value of static pressure at the bottom wall against number of grid elements.

2.5 Validation

The validation of the current numerical methodology is conducted for both the cases i.e. non-reacting and cold flow with H_2-injection by comparing the computed results with the experimental and computational data reported in the literature. Quantitative analysis is performed for the non-reacting case, by comparing static pressure variation along the lower and upper walls of the combustor with respect to Overmann's computational [17] and Waidmann's experimental results [18], as shown in Fig. 4(a). The computed static pressure distribution along the centerline of the engine is compared with that from the Overmann's computational and experimental results and plotted in Fig. 4(b). The wedge-shape strut, acting as an obstruction to the supersonic flow, generates oblique shocks at its leading edge (L.E.) and expansion waves at its trailing edge. The formation of the shock at L.E., which is reflected off from the upper and lower walls of the combustor, causes an increase in static pressure near these walls. However, the interaction of expansion fans leads to a reduction in static pressure. Further along the stream, the static pressure increases again due to the interaction between reflected shocks and oblique shocks.

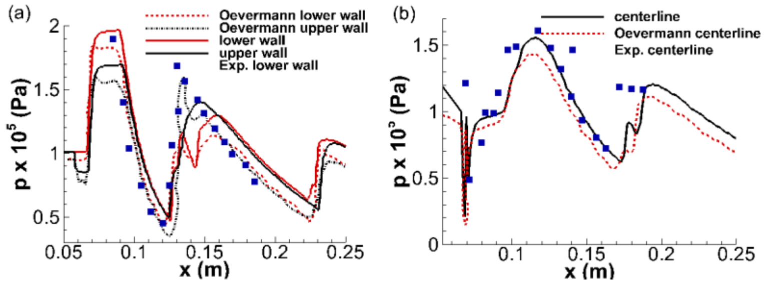

Fig. 4 Comparison of static pressure variation along a) lower wall and upper wall and b) centerline, with respect to computational [18] and experimental results [19].

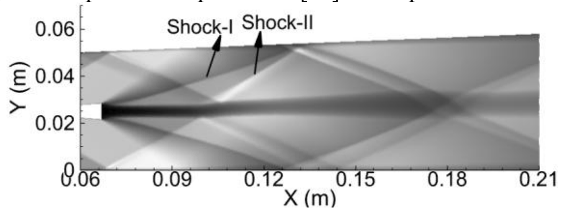

Fig. 5 Simulated density contours showing shock structures as shock-I and shock-II.

Figure 5 shows the simulated density contours within the scramjet combustor for the cold flow case. It is observed that the numerical methodology captures the shock structures labeled as shock-1 and shock-2, consistent with the experimental results [19], thus validating the present numerical methodology qualitatively. The cross streamwise velocity profiles at four different streamwise locations; *x = 78mm, 125mm, 157mm* and *233mm* are plotted in Fig. 5 and compared with the computational [17] and experimental [18] data. The single peak in experiment as contrast to double peak in velocity for present case at x = 78 mm could be due to three-dimensional effect for the experiment. Once the H_2 jet exits the wedge at 1200 m/s, it is quickly accelerated in the low-pressure recirculation zone behind the base of the strut achieving maximum velocity of 2100 m/sec. It is then decelerated by the shock wave to the velocity of 400 m/sec which is one-third of the jet exit velocity at around x = 78 mm. Thereafter, the jet gets accelerated to a constant velocity of 700 m/sec at around x = 157 mm. From all the frames of Fig. 6, it is observed that the computed minimum velocity inside the H_2 jet is shifted upwards as compared to Oevermann's CFD results [17] and experimental measurements of Waidmann [18]. Overall, the computed cross-streamwise velocity profile matches reasonably well (both quantitatively and qualitatively) with the results reported in literature [17,18].

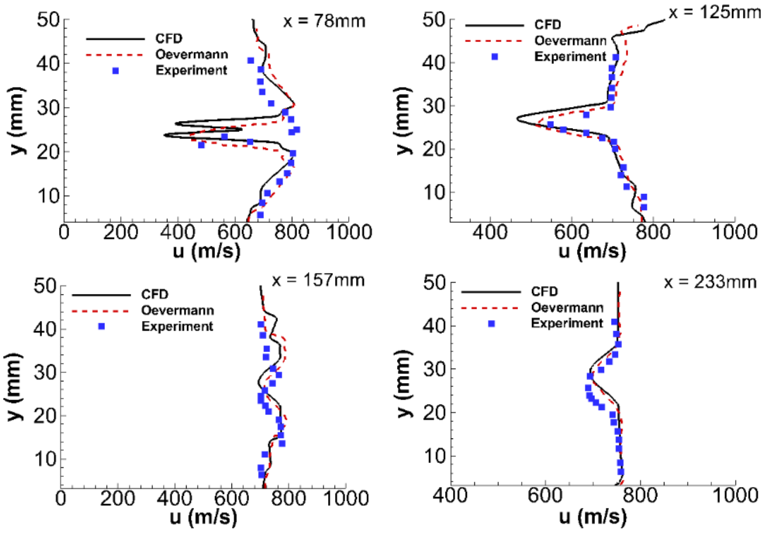

Fig. 6 Variation of cross-streamwise velocity at four streamwise locations of x = 78mm, 125 mm, 157 mm and 233 mm, compared with the values reported in literature [17,18].

The reacting flow case is performed by comparing the computed axial velocity variation along the combustor's centerline ($y = 25\, mm$) with respect to that from the Oevermann's CFD data [18] and experimental data of Waidmann's [19], as plotted in Fig. 7. One can notice that initially the velocity reduces across the strong oblique shock then at around $x \sim 100$ mm increases across the expansion fan and then decreases again at around $x \sim 100$ mm due to the weak shock. Overall, the computed axial velocity profile matches reasonably well with that from the computational [17] and experimental [18] results.

3 Results and Discussions

3.1 Impact of Injector Configuration on Combustion Characteristics

The flow characteristics of the reacting flow case for the modified scramjet combustor model, featuring three different fuel injection strut designs, are analyzed. Contours of Mach number for the wedge, lobed and diamond struts are plotted in Fig. 8. It is observed that an oblique shock of equal strength forms at the leading edge of all three struts, as the wedge angle is consistent across the three strut designs. However, the expansion shock near the base of the strut is strongest for the lobed strut, followed by the diamond, and weakest for the wedge strut. It is observed that the Mach number decreases in the middle of the scramjet for all three strut configurations due to the interaction of reflected shocks with the oblique shocks. Several studies have shown that vortex, turbulence and shock waves play a crucial role in the effective blending of air and H_2 within the combustor. Therefore, the z-component of vorticity (ω_z) contours are plotted for the wedge, lobed and diamond strut designs in Fig. 9 along with streamlines illustrating the flow direction. One can observe that the current numerical simulations accurately capture the recirculation zone behind the strut base. Positive (blue) values and negative (red) values of z-vorticity (ω_z) indicates the clockwise and counterclockwise rotating vortices in the recirculation zone, respectively. It is observed that the H_2 injection divides the recirculation region into two zones: one above and one below the fuel injection point. Additionally, the recirculation zone is strongest for the lobed strut with maximum (ω_z) magnitude, followed by the wedge. Although the vorticity is significant for the diamond strut, the recirculation zone is weak or nearly nonexistent.

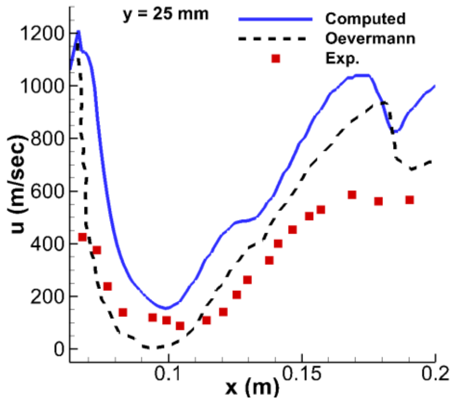

Fig. 7 Computed axial velocity profile of reacting flow at y = 25mm with literature [17,18].

3.2 Performance Parameters

The efficacy of the modified scramjet engine model with the wedge, lobed and diamond strut designs is investigated by evaluating the combustion efficiency (η_c) and total pressure loss (η_{tpl}) [17, 18]. Figure 10 illustrates the (a) combustion efficiency and (b) total pressure loss for wedge, lobed and diamond struts with respect to the length of the combustor. The x-axis of the η_c plot starts just after the strut due to the lack of fuel upstream. After a short distance, η_c increases till the end of the combustor, indicating that the H_2-air mixing significantly impacts the combustion efficiency. It is observed that the diamond strut leads to inadequate and weaker combustion inside the combustor.

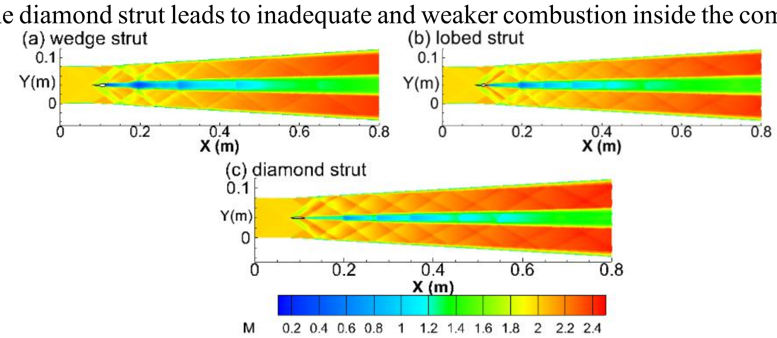

Fig. 8 Mach contours for the reacting case for (a) wedge, (b) lobed and (c) diamond struts.

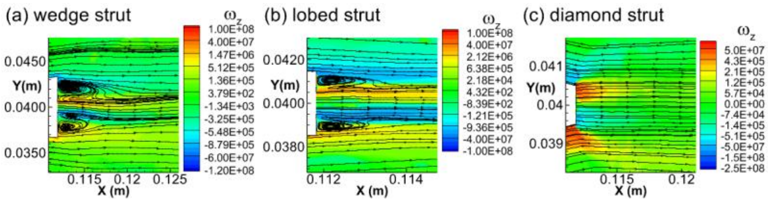

Fig. 9 z-vorticity (ω_z) contours for the reacting case for (a) wedge, (b) lobed and (c) diamond.

Here, wedge strut with $\eta_c \cong 69\%$ has the highest efficiency, lobed has slightly reduced $\eta_c \cong 67\%$ and diamond strut with least $\eta_c \cong 58\%$ among the three strut designs analyzed. Pressure loss occurs due to mixing and ignition just downstream of the injection. It is noticed that the total pressure loss is highest for the lobed strut and lowest for the diamond strut. This is due to the non-existence of recirculation zone implying less mixing and weaker combustion thus resulting in least pressure loss for the case of diamond strut. However, the strong and wide recirculation zones created by the lobed strut led to the highest total pressure loss.

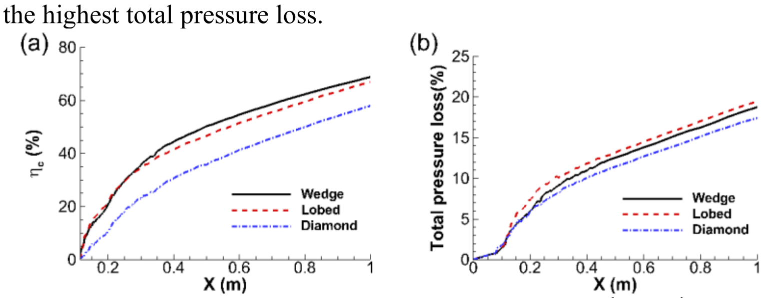

Fig. 10 (a) Combustion efficiency (η_c) and (b) total pressure loss (η_{tploss}) for wedge (solid line), lobed (dashed line) and diamond (dash dotted line).

4 Summary and Conclusion

In the current study, the scramjet engine performance is assessed by evaluating the combustion efficiency (η_c) and the total pressure loss (η_{ptl}) for the wedge, lobed and diamond strut geometries. Numerical simulations are performed by solving the steady RANS equations with the SST *k-ω* turbulence model. The mesh independence study yielded the optimal grid for which results are independent of grid resolution. The numerical methodology is validated qualitatively and quantitatively for the single wedge strut in a traditional DLR combustor for both the reacting and cold flow with H_2 injection cases. The findings show strong recirculation at the base of the wedge and lobed struts, but none for the diamond strut. This led to the highest η_c (69%) for the wedge, followed by the lobed strut at 67%, with the diamond strut having the lowest η_c. Moreover, the total pressure loss is highest for the lobed strut followed by wedge and least for the diamond strut. This research will serve as a guide for researchers and industries seeking optimal fuel strut geometry for scramjet combustors.

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